Muhammad Haseeb **Arshad**

Electrical Engineer | Power Electronics, Machine Learning

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125 Gulmuhar Block, Bahria Town, Lahore - Pakistan
Born on October 10, 1989

Skills

Programming	**C++**, **MATLAB**, **Python**, VHDL, MAPLE, R
Engineering Softwares	**MATLAB**, **Proteus**, **NI MultiSim**, **CorelDraw**, NI LABView, Kiel, Photoshop CS5
Operating System	**Windows 10**, **Mac OS X**, Windows 7, Linux Redhat
Other	RStudio, Jupitor Notebook, AVR Studio, dSpace, Active-HDL, MS Office, MS Visio, CNN, RNN, LSTM, VAEs

Education

2018-2020 — **M.Sc. Electrical Engineering (Specialization: Control & System Engineering)**
- King Fahd University of Petroleum & Minerals, Dhahran, Saudi Arabia
- CGPA 3.857/4.0

Thesis: *Intelligent Model Predictive Torque Control of Induction Motor*

2016-2017 — **M.Sc. Electrical Engineering (Specialization: Power & Control Systems)**
- University of Engineering & Technology, Lahore, Pakistan
- CGPA 3.63/4.0

Design Project: *Advance Direct Torque Control of Two Phase Symmetrical Induction Motor fed with B4 Inverter*

2009-2013 — **B.Sc. Electrical Engineering (Specialization: Electronics & Telecommunication)**
- University of the Punjab, Lahore, Pakistan
- CGPA 3.81/4.0

Thesis: *Wandering Autonomous Lifelike Behaviour Based Ground Vehicle*

Professional Training & Courses

21 June 2020 – 25 June 2020 — **Data Science Fundamentals, KFUPM, KnowledgeHut**
- Data Science in Businees
- Data Analytics Tools
- Data Science Methodology

Zoom | Jupitor Notebook

September – December 2019 — **ICS-460 | Introduction to Machine Learning, KFUPM, Saudi Arabia**
- Machine Learning and its Fundamentals
- Supervised, Unsupervised & Reinforced Learning
- Regression vs Classification Techniques
- Introduction to Deep Neural Networks
- Implementation of Machine Learning Algorithms using Jupitor Notebook

Jupitor Notebook | MATLAB | Excel

July 2017 – September 2017 — **The Data Scientist's Toolbox | Data Science Specialization, John Happkins University, Coursera**
- Introduction to Data Science
- Big Data Analytics

MS Word

Work Experience

May 2020 – September 2018 — **Research Assistant, KFUPM, Saudi Arabia**
FACTS LAB for Department of Electrical Engineering
Graduate Researcher under Dr. Mohammed Ali Abido, working on
- Optimization of FCS-MPTC for Induction Motor
- Investigating the performance of PCC for line-start PMSM

MATLAB | MultiSim | Simulink | dSpace | RTDS | LaTeX

December 2019
September 2019

Teacher Assistant, KFUPM, Saudi Arabia
EE-460 Power Electronics for Department of Electrical Engineering
Grader & Summary Class Instructor
- Assignment and quiz grader
- Project groups co-supervisor

MATLAB | MultiSim | PSpice | MAPLE | LaTeX

May 2019
April 2019

Teacher Assistant, KFUPM, Saudi Arabia
Assitant to Dr. Umar Johar for Department of Electrical Engineering
Personal Assistant for Clerical Work
- Book Keeping of student records of past three years (2016-2019)
- Helper for academic petition forms

Excel | MS Word | Outlook

Aritcles under Review

Genetic Algorithm tuned Adaptive Discrete-time Sliding Mode Controller for LCL Grid-Connected Inverter - Not published yet February 2021

Submitted to Journal of the Franklin Institute

MPC of LSPMSM Using A Novel Flux Observer Based on Indirect Current Method - Not published yet January 2021

Submitted to IEEE Access

Punlished Aritcles

Advanced Direct Torque Control of Four Switch Fed Two-Phase Symmetric Induction Motor June 2018

Link to Publication

A Chaos Based SVPWM technique for B4 Inverter fed Two-Phase Symmetric Induction Motor for THD & EMI improvement at Low Modulation Index July 2019

Link to Publication

Weighting Factors Optimization of Model Predictive Torque Control of Induction Motor Using NSGA-II With TOPSIS Decision Making December 2019

Link to Publication

An Overview of Sequential Learning Algorithms for Single Hidden Layer Networks: Current Issues & Future Trends June 2020

Link to Publication

Artificial Bee Colony Optimized Self-tuning PI Speed Controller for FCS-MPCC of Permanent Magnet Synchronous Machines December 2020

Link to Publication

A Simple Technique for Studying Chaos using Jerk Equation with Discrete Time Sine Map January 2021

Link to Publication

Hierarchical Control of DC Motor coupled with Cuk Converter combining Differential Flatness and Sliding Mode Control January 2021

Link to Publication

Important Coursework

- Convex Optimization
- Evolutionary Optimization
- Intelligent Control
- Adaptive Control
- Stochastic Process
- Machine Learning
- Digital Signal Processing
- Linear Control Systems
- Advance Power Electronics
- Design of Digital System

Achievements

- Got 1st position in the of Punjab Group Colleges during the session 2006-08
- G ot 4th position in B.Sc. Electrical Engineering at university level during the session 2009-13
- HEC Indigenous Scholar for my postgraduate studies at UET Lahore
- Got 4th position in M.Sc. Electrical Engineering at university level during the session 2016-17
- Graduate Scholar at KFUPM, Dhahran, Saudi Arabia
- Won Travel Grant from IES for ISIE 2018 conference.

Languages

Urdu	●●●●●
Punjabi	●●●●●
English	●●●●○

Forces

- Passionate
- Motivated
- Autonomous

References

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